Chapter 5. The C++ memory model and operations on atomic types

5.1. Memory model basics

5.1.1. Objects and memory locations

5.1.2. Objects, memory locations, and concurrency

5.1.3. Modification orders

5.2. Atomic operations and types in C++

5.2.1. The standard atomic types

5.2.2. Operations on std::atomic_flag

5.2.3. Operations on std::atomic<bool>

5.2.4. Operations on std::atomic<T*>: pointer arithmetic

5.2.5. Operations on standard atomic integral types

5.2.6. The std::atomic<> primary class template

5.2.7. Free functions for atomic operations

5.3. Synchronizing operations and enforcing ordering

5.3.1. The synchronizes-with relationship

5.3.2. The happens-before relationship

5.3.3. Memory ordering for atomic operations

5.3.4. Release sequences and synchronizes-with

5.3.4. Fences

5.3.6. Ordering nonatomic operations with atomics

5.4. Summary

Chapter 6. Designing lock-based concurrent data structures

6.1. What does it mean to design for concurrency?

6.1.1. Guidelines for designing data structures for concurrency